

Unit 9: Second-Order Linear Equations

Prepared by Staples Education

Condensed Cheat Sheet: Theory and Structure

Unit 9 at a Glance: Core Sub-Topics

A second-order linear equation reads $a(x)y'' + b(x)y' + c(x)y = g(x)$; this unit builds the homogeneous theory and the constant-coefficient solution machine.

- **Existence, uniqueness, superposition** — two side conditions pin a unique solution; linear combinations of solutions are solutions.
- **The Wronskian** — $W(y_1, y_2) = y_1y_2' - y_2y_1'$ tests linear independence and certifies a fundamental set.
- **Abel's identity** — the Wronskian itself solves a first-order ODE, so it is either never zero or identically zero.
- **Reduction of order** — one known solution y_1 manufactures a second, independent y_2 .
- **Constant-coefficient roots** — the characteristic equation $ar^2 + br + c = 0$ splits into real-distinct, repeated, and complex cases.
- **Cauchy–Euler and higher order** — the x^m trick and the degree- n characteristic polynomial.

Part I — Condensed Cheat Sheet

9.1 Introduction to Second-Order Equations

The general linear second-order equation is $a(x)y'' + b(x)y' + c(x)y = g(x)$. It is **homogeneous** when $g(x) = 0$ and **nonhomogeneous** otherwise; its general solution carries *two* arbitrary constants.

Standard and Normalized Forms

$$a(x)y'' + b(x)y' + c(x)y = g(x), \quad y'' + p(x)y' + q(x)y = r(x),$$

with $p = b/a$, $q = c/a$, $r = g/a$ where $a(x) \neq 0$. Linearity means y , y' , y'' each appear to the first power and are never multiplied together.

9.2 Existence, Uniqueness, Superposition, and Independence

On any interval where p , q , and r are continuous, the initial value problem with $y(x_0) = y_0$, $y'(x_0) = y_0'$ has a *unique* solution.

Superposition Principle

If y_1 and y_2 each solve the homogeneous equation $y'' + p y' + q y = 0$, then for *any* constants c_1, c_2 ,

$$y = c_1 y_1 + c_2 y_2$$

is also a solution. The general solution is this two-parameter family precisely when y_1, y_2 are linearly independent.

9.3 Linear Independence and the Wronskian

Two solutions form a **fundamental set** exactly when they are linearly independent, and the Wronskian is the determinant that tests it.

The Wronskian

$$W(y_1, y_2)(x) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1'.$$

If $W(x_0) \neq 0$ at any point, y_1, y_2 are **linearly independent** on the interval and $\{y_1, y_2\}$ is a fundamental set.

9.4 Abel's Identity

Abel's identity says the Wronskian of two solutions of $y'' + p(x)y' + q(x)y = 0$ itself satisfies a first-order ODE, so its sign can never change.

Abel's Formula

For $y'' + p(x)y' + q(x)y = 0$,

$$W(x) = W(x_0) \exp\left(-\int_{x_0}^x p(t) dt\right).$$

Because the exponential is never zero, W is **either identically zero** (dependent solutions) **or never zero** (independent) — there is no middle ground.

9.5 Reduction of Order

Given one nonzero solution y_1 of the homogeneous equation, reduction of order substitutes $y_2 = v(x)y_1$ and solves for v , producing a second independent solution.

Reduction of Order Formula

For $y'' + p(x)y' + q(x)y = 0$ with a known solution y_1 ,

$$y_2 = y_1(x) \int \frac{e^{-\int p(x) dx}}{y_1(x)^2} dx.$$

The substitution $y_2 = v y_1$ collapses the second-order equation to a *first-order* equation in v' , which the integral above solves directly.

9.6 Constant Coefficient Homogeneous Equations

For $ay'' + by' + cy = 0$ with constants a, b, c , try $y = e^{rx}$ to get the **characteristic equation** $ar^2 + br + c = 0$. The discriminant $b^2 - 4ac$ selects one of three cases.

The Three Root Cases

$$ar^2 + br + c = 0, \quad r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

- **Real distinct** ($b^2 - 4ac > 0$), roots $r_1 \neq r_2$:

$$y = c_1 e^{r_1 x} + c_2 e^{r_2 x}.$$

- **Repeated** ($b^2 - 4ac = 0$), one root r :

$$y = (c_1 + c_2 x) e^{rx}.$$

- **Complex conjugate** ($b^2 - 4ac < 0$), $r = \alpha \pm \beta i$:

$$y = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x).$$

Common Pitfall: The Repeated-Root Boundary

The repeated case ($b^2 - 4ac = 0$) is the boundary between real and complex roots. Do *not* write $y = (c_1 + c_2)e^{rx}$ — that is a single constant; the second solution carries the factor x , giving $y = (c_1 + c_2 x)e^{rx}$. For complex roots, α is the *real* part (in the exponential) and β the *imaginary* part (inside the sine and cosine); swapping them is the classic sign trap.

9.7 The Cauchy–Euler Equation

The Cauchy–Euler (equidimensional) equation $ax^2y'' + bxy' + cy = 0$ has powers of x matching derivative orders. Try $y = x^m$.

Cauchy–Euler Root Cases

Substituting $y = x^m$ gives the indicial equation $am(m-1) + bm + c = 0$:

- **Distinct** $m_1 \neq m_2$: $y = c_1 x^{m_1} + c_2 x^{m_2}$.
- **Repeated** m : $y = (c_1 + c_2 \ln x) x^m$.
- **Complex** $m = \alpha \pm \beta i$:

$$y = x^\alpha (c_1 \cos(\beta \ln x) + c_2 \sin(\beta \ln x)).$$

9.8 Higher-Order Linear Equations

An n th-order constant-coefficient equation has a degree- n characteristic polynomial; each real root contributes an exponential, and a root of **multiplicity** k contributes k independent terms.

Repeated Roots at Higher Order

A root r of multiplicity k contributes

$$e^{rx}, xe^{rx}, x^2e^{rx}, \dots, x^{k-1}e^{rx},$$

and a complex pair $\alpha \pm \beta i$ of multiplicity k contributes the analogous $x^j e^{\alpha x} \cos \beta x$ and $x^j e^{\alpha x} \sin \beta x$ for $j = 0, \dots, k-1$.

Unit 9 Key Takeaway

The general solution of a homogeneous second-order linear equation is $c_1 y_1 + c_2 y_2$ for any **linearly independent** pair, which the **Wronskian** certifies and **Abel's identity** guards. For constant coefficients the **characteristic roots** dictate the form — **real-distinct**, **repeated** (carry the x), or **complex** ($e^{\alpha x}$ times sines and cosines) — and **reduction of order** recovers a second solution from a first.